

# U.S. Payer API, Pricing & Payer-Side Opportunity

## A Two-Sided Feasibility Study

Provider / RCM workflows · Payer-side claims intelligence · API pricing · Payer guidelines · Payment integrity · Regulatory models

PREPARED FOR      **AI Revenue Recovery OS**

DATE      August 18, 2026

SCOPE      Can we start provider-side on existing APIs/clearinghouses, and become a payer-side claims intelligence & operations platform? Data availability, source, standard, pricing, enrollment, and legal/operational boundaries.

CLASSIFICATION      Confidential · Investor & product/engineering feasibility · Not legal advice

### Disclosures

Evidence-backed from official CMS / X12 / HL7 / CAQH / eCFR / NAIC and vendor documentation (research August 2026). Values are YES / NO / PARTIAL / UNKNOWN / VARIABLE. Free sandbox is never equated with free production; direct payer API never with clearinghouse connectivity. No pricing is invented. Payer-side sections describe the regulatory landscape and are **not legal advice** — Models B/C require regulatory/legal review. No PHI.

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# 1. Executive Summary

This study extends the provider-side payer-API feasibility analysis to a **two-sided** question: can AI Revenue Recovery OS begin as a provider/RCM platform on existing APIs and clearinghouses, and evolve into a payer-side claims-intelligence and operations platform? The short answer is **yes, with conditions**.

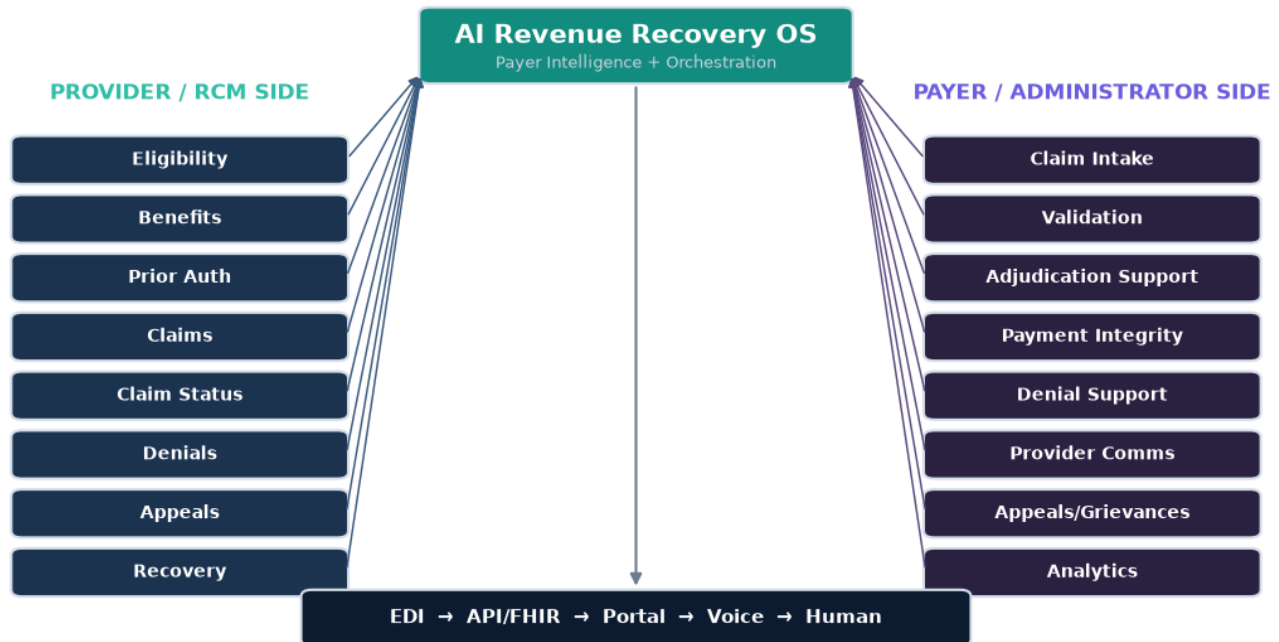


Figure 1. The two-sided model — one orchestration + payer-intelligence core serving provider-side RCM workflows and payer-side claims operations.

## Final answer — YES, WITH CONDITIONS

Provider-side is buildable now on clearinghouse APIs, with a near-\$0 MVP via free sandboxes and Stedi's free production tier. The payer side is a real, large opportunity as **Model A** (SaaS decision-support) and **Model B** (administrative services); **Model C** (delegated claims/UM/payment decisions) is a long-term, heavily-regulated frontier. Across every model, **final medical-necessity / coverage decisions cannot be made by AI alone** — a qualified clinician must own any adverse determination (CMS-4201-F for Medicare Advantage; California SB 1120).

## 2. Research Methodology

Primary sources first: CMS (rules, fact sheets, appeals, Medicare Coverage Database), eCFR (Titles 42/29/45), NAIC (TPA model), URAC/NCQA (UM accreditation), ASC X12, HL7 (FHIR / Da Vinci / CARIN), CAQH, and vendor developer/pricing documentation. Trade press only corroborates figures.

- Values are YES / NO / PARTIAL / UNKNOWN / **VARIABLE** (used for contract/plan/state-dependent payer rules). Never forced binaries.
- Distinctions preserved: standard vs electronic vs API vs public vs free-sandbox vs free-production vs paid; and direct payer API vs clearinghouse connectivity.
- Scores (payer-side opportunity, automation readiness) are *our* analytical constructs, not industry standards.
- Payer-side legal/regulatory content is a sourced landscape overview, **not legal advice**; it flags where regulatory/legal review is required.
- Coverage limitation: 43 payer organizations are representative, not exhaustive; open items are enumerated in the Research Gaps appendix.

### 3. Provider-Side Workflows

The provider-side foundation is established (see the companion feasibility report for the full 113-field payer matrix). In summary, the standardized transactions are reachable today via clearinghouse APIs; prior authorization and appeals are the human-heavy frontier.

| Workflow               | Standard        | API today                | Free path            | Build status |
|------------------------|-----------------|--------------------------|----------------------|--------------|
| Eligibility            | 270/271         | YES (clearinghouse)      | Stedi free tier      | GREEN        |
| Benefits               | 271 (CORE)      | YES                      | Stedi/Optum          | GREEN        |
| Claim Status           | 276/277         | YES                      | Stedi free tier      | GREEN        |
| Claims (837)           | 837P/I/D        | YES                      | Stedi/Optum/Availity | GREEN        |
| Denials (835)          | 835 + CARC/RARC | YES                      | Stedi (enroll)       | GREEN        |
| Patient Responsibility | 271 + calc      | PARTIAL (own engine)     | N/A                  | YELLOW       |
| Prior Authorization    | 278 / FHIR PAS  | PARTIAL (2027 FHIR)      | Availity/portal      | ORANGE       |
| Appeals                | (no standard)   | PARTIAL (payer-specific) | portal/fax/mail      | RED          |

## 4. API Landscape

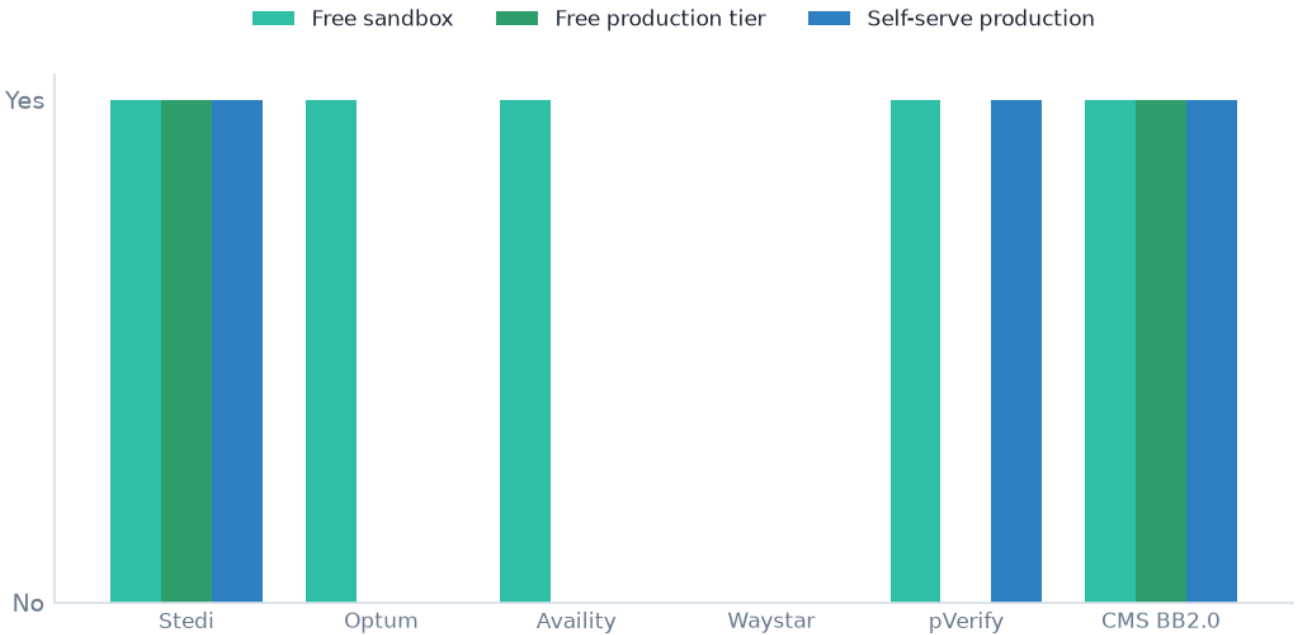
The per-API inventory (Excel: API MASTER) records each API separately with its standard, direct-vs-intermediary status, sandbox/production, pricing and enrollment. The pattern: connectivity is intermediated by clearinghouses; direct free provider-facing payer APIs are the exception.

| Provider       | API                                    | Workflow                      | Standard                | Type    | Free Sandbox | Free Prod    | Conf   |
|----------------|----------------------------------------|-------------------------------|-------------------------|---------|--------------|--------------|--------|
| Stedi          | Eligibility API                        | Eligibility                   | 270/271                 | Interm. | YES          | YES (100/mo) | High   |
| Stedi          | Claim Status API                       | Claim Status                  | 276/277                 | Interm. | YES          | YES (100/mo) | High   |
| Stedi          | Claims API (837)                       | Claims                        | 837P/I/D                | Interm. | YES          | YES (100/mo) | High   |
| Stedi          | Remittance (835) API                   | Denials                       | 835                     | Interm. | YES          | YES (100/mo) | High   |
| Availity       | Eligibility & Benefits API             | Eligibility/Benefits          | 270/271                 | Interm. | YES          | NO           | High   |
| Availity       | Claim Status API                       | Claim Status                  | 276/277                 | Interm. | YES          | NO           | High   |
| Availity       | Authorization/Referral (278) + FHIR PA | Prior Auth                    | 278 / FHIR PAS          | Interm. | YES          | NO           | High   |
| Optum (Change) | Eligibility API                        | Eligibility                   | 270/271                 | Interm. | YES          | NO           | High   |
| Optum (Change) | Claim Status API                       | Claim Status                  | 276/277                 | Interm. | YES          | NO           | High   |
| Optum (Change) | Claims + ERA + Attachments (275)       | Claims/Denials                | 837/835/275             | Interm. | YES          | NO           | High   |
| Waystar        | Eligibility/Status/Claims/ERA          | Multiple                      | 270/271·276/277·837·835 | Interm. | UNKNOWN      | NO           | Medium |
| Zelis          | ERA/EFT enrollment + delivery          | Denials/Payment               | 835                     | Interm. | YES          | NO           | High   |
| pVerify        | Eligibility/Status/PA API              | Eligibility/Estimation        | 270/271 + FHIR          | Interm. | YES          | NO           | Medium |
| CMS            | Blue Button 2.0                        | Patient claims (Medicare FFS) | FHIR R4 (CARIN)         | Direct  | YES          | YES          | High   |

Full inventory (18 APIs incl. CMS Blue Button 2.0, DPC, AB2D, NPPES and payer Patient/Provider Access FHIR) is in the workbook.

## 5. Pricing Landscape

The investor question — are the APIs free? — resolves precisely: free sandboxes are common; a free production tier is rare (Stedi, and CMS's patient-authorized Medicare APIs). Optum and Availity gate production behind contracts; Waystar is enterprise. No per-transaction price is invented — where sales-gated, it is recorded 'not publicly disclosed'.



CMS Blue Button 2.0 = free but Medicare-FFS, patient-authorized (not general RCM)

Figure 2. Free sandbox vs free production vs self-serve production across leading API providers.

**Do not conflate**

'Free sandbox' ≠ 'free production API'. 'Clearinghouse supports payer X' ≠ 'payer X provides an API'. 'Patient Access FHIR' ≠ 'provider RCM feed'. Each is a separate field in the database.

## 6. Free MVP API Stack

A genuine \$0 build/test MVP is achievable by combining synthetic data, public FHIR test servers, CMS sandboxes and free provider/coverage data. Real production payer transactions still require enrollment and — beyond Stedi's free tier — usage fees.

| Resource                               | Provider      | Free?         | Sandbox/Production                              | Limits                                                        |
|----------------------------------------|---------------|---------------|-------------------------------------------------|---------------------------------------------------------------|
| Blue Button 2.0                        | CMS           | YES           | Sandbox (synthetic) + prod (app review)         | 10k synthetic users; ~4 yrs history; 60M+ beneficiaries       |
| Data at the Point of Care              | CMS           | YES           | Sandbox open; prod pilot (apps paused — verify) | Provider-treatment relationship; production onboarding paused |
| AB2D                                   | CMS           | YES           | Public sandbox; prod gated to PDP sponsors      | Only active PDP sponsors                                      |
| Synthea / SyntheticMass                | MITRE         | YES           | Generate locally / download                     | Synthetic only; SyntheticMass API is STU3                     |
| HAPI FHIR public server                | HAPI/Smile    | YES           | Public test server                              | Data periodically wiped; no SLA; not for PHI                  |
| Da Vinci reference implementations     | HL7 Da Vinci  | YES           | Public RIs / self-host                          | Demos, not production payers                                  |
| Logica / Meld sandbox                  | Logica Health | YES           | Sandbox                                         | Retiring to Meld CE — verify current status                   |
| Inferno / ONC (g)(10)                  | ONC / HHS     | YES           | Hosted + self-host                              | Testing tool, not a data source                               |
| NPPES NPI Registry API                 | CMS/HHS       | YES           | Public production, no token                     | Undocumented rate limits; ~200 results/req                    |
| CMS datasets (PDC, MCD, fee schedules) | CMS           | YES           | Public download / data.cms.gov                  | Reference/bulk, not real-time claims                          |
| Stedi (free tier)                      | Stedi         | YES           | Sandbox + free Basic production                 | 100 free/mo each; then pay-as-you-go                          |
| Availity (Demo)                        | Availity      | YES (sandbox) | Demo (mock); production gated                   | No PHI in demo; production application required               |
| Optum (sandbox)                        | Optum         | YES (sandbox) | Sandbox; production paid subscription           | No production-data testing in sandbox                         |
| CARIN Blue Button RI                   | HL7/CARIN     | YES           | Reference implementation / self-host            | RI + samples, not a live payer                                |

A genuine \$0 build/test MVP is achievable (Synthea + HAPI/Da Vinci/CARIN RIs + Inferno + NPPES/CMS data + CMS Blue Button/DPC/AB2D sandboxes). Real production payer transactions generally require payer enrollment and — beyond Stedi's free Basic tier (100/mo each) — usage-based clearinghouse fees. Verify current status of DPC (production paused) and Logica (retiring to Meld).



## 7. Production API Stack

Recommended production connectivity per capability. The principle: one clearinghouse integration reaches thousands of payers; add payer FHIR selectively (2027 Provider Access), and keep portal/voice/human as fallback.

| Capability               | Recommended (why)                        | Fallback                   |
|--------------------------|------------------------------------------|----------------------------|
| Eligibility / Benefits   | Stedi (self-serve, free tier, JSON+X12)  | Optum · Availity           |
| Claim Status             | Stedi (real-time)                        | Optum · portal · voice     |
| Claims (837)             | Stedi / Optum (broad reach)              | Availity                   |
| ERA / Denials (835)      | Stedi (+ per-payer enrollment)           | Zelis (payments)           |
| Prior Authorization      | Availity (278 + FHIR PA); payer portals  | Voice; FHIR PAS (2027)     |
| FHIR / member data       | Payer Patient Access; CMS Blue Button    | Provider Access API (2027) |
| Attachments              | Optum (275)                              | Da Vinci CDex (emerging)   |
| Broad payer connectivity | Optum/Change (widest) + Availity (Blues) | Waystar (enterprise)       |

## 8. Payer-Side Opportunity

Beyond serving providers and RCM BPOs, AI Revenue Recovery OS could serve **payers**. We scored ten payer-side workflows on market, feasibility, API availability, regulatory complexity, integration, human dependency, AI opportunity and revenue potential. The sweet spot is **low-regulatory, high-AI** work — claim intake/validation, payment integrity, and provider communication — delivered first as decision-support (Model A).

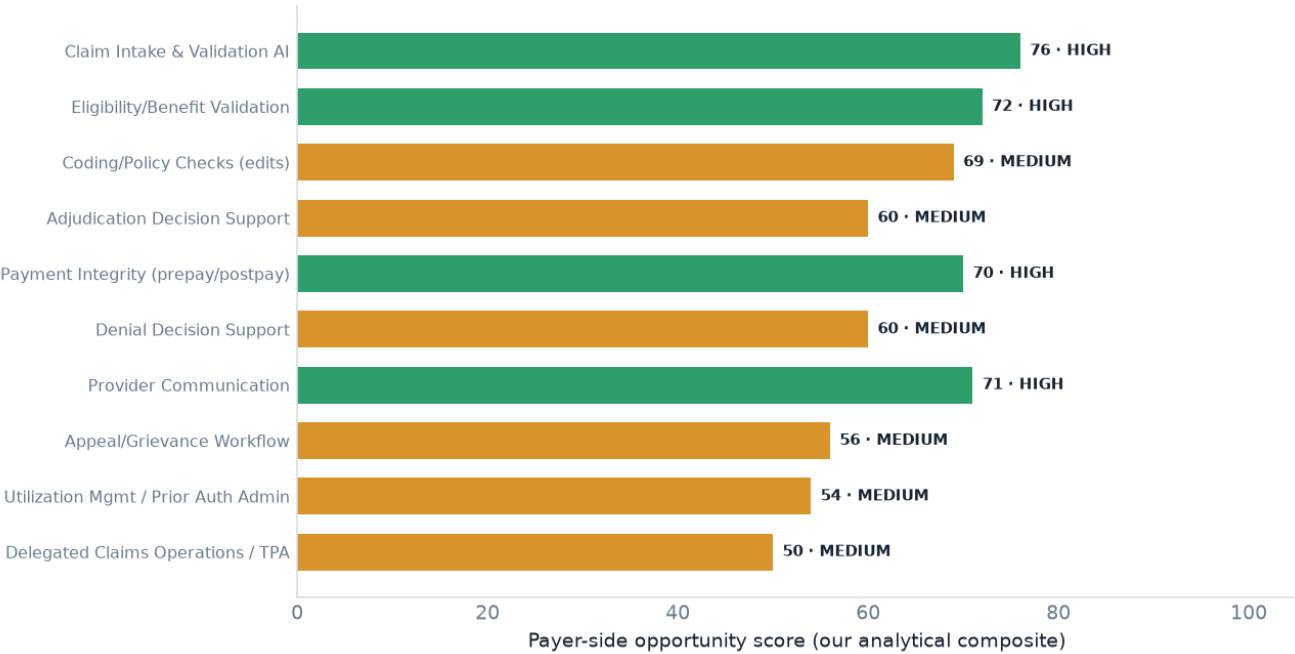


Figure 3. Payer-side opportunity scores (our analytical composite). Green = HIGH, amber = MEDIUM, red = LOW.



Figure 4. Regulatory complexity vs AI opportunity (bubble = revenue potential). Start where regulation is low and AI leverage is high.

## 9. Payment Integrity

Payment integrity — prepay/postpay review, overpayment/duplicate detection, DRG validation, coordination-of-benefits and fraud/waste/abuse — is the highest-scoring payer-side wedge, and a large, consolidating market. A new entrant needs a specific wedge (a claim type, clean prepay integration, or an orchestration/operating-system play), not head-on breadth.

| Vendor                   | Category                                  | Orientation  |
|--------------------------|-------------------------------------------|--------------|
| Cotiviti                 | Payment accuracy (prepay+postpay)         | Payer        |
| Optum (Optum Insight)    | Payment integrity across claim lifecycle  | Payer        |
| Zelis                    | Claims pricing + payment integrity        | Payer        |
| EXL Health               | Payment integrity analytics + services    | Payer        |
| Gainwell (incl. HMS)     | Government/Medicaid payment integrity     | Payer (govt) |
| Lyric (ex-ClaimsXten)    | Prepay claims editing / payment accuracy  | Payer        |
| ClarisHealth             | Payment-integrity operating system (SaaS) | Payer        |
| Machinify (New Mountain) | AI-native payment integrity               | Payer        |
| Codoxo                   | GenAI FWA + payment integrity             | Payer        |

U.S. payment-integrity market-size estimates vary widely by definition (software-only vs services-inclusive): e.g., Precedence Research ~\$10.3B by 2033; other firms range ~\$1–15B across 2024–2025 bases. Cite a range, not a point — Low confidence.

# 10. Competitive Landscape

The landscape splits by orientation: payer-side decisioning (Cohere, Anterior), provider-side submission (Develop Health, Myndshft), connecting networks (Availity, Rhyme), and provider-side RCM AI (Waystar, Adonis, Candid, Infinitus, Fathom). CMS-0057-F FHIR prior-auth (2027) is a tailwind shaping all of them.

| Company                              | Category                      | Side     |
|--------------------------------------|-------------------------------|----------|
| Cohere Health                        | Payer UM / prior auth AI      | Payer    |
| Availity (AuthAI)                    | Clearinghouse + auth AI       | Both     |
| Rhyme (ex-PriorAuthNow)              | EHR-integrated PA network     | Both     |
| Myndshft (DrFirst)                   | PA + benefits verification    | Provider |
| Anterior (ex-Co:Helm)                | Payer clinical decisioning AI | Payer    |
| Develop Health                       | GenAI medication-access / PA  | Provider |
| MCG / InterQual                      | UM clinical criteria          | Payer    |
| Waystar                              | End-to-end RCM software       | Provider |
| Adonis                               | AI RCM orchestration          | Provider |
| Candid Health                        | Autonomous RCM / billing      | Provider |
| Smarter Technologies (Thoughtful AI) | Enterprise AI RCM             | Provider |
| Infinitus                            | Agentic voice AI              | Provider |
| Fathom                               | Autonomous medical coding     | Provider |

## 11. Payer Guidelines

Provider guidelines — timely filing, appeal deadlines and submission methods — are a critical operational input. The key finding: **commercial windows are contract/plan/state-variable** (documented range ~90 days to 12–18 months) and must be stored as configurable values, never hard-coded. Only the government frameworks are fixed rules.

| Payer                 | Timely Filing                                                    | Appeal Deadline                                                                 | Method                                         | Conf         |
|-----------------------|------------------------------------------------------------------|---------------------------------------------------------------------------------|------------------------------------------------|--------------|
| UnitedHealthcare      | VARIABLE (per participation agreement; MA claims 365d floor)     | VARIABLE by LOB (~65d MA reconsideration)                                       | Provider portal / Availity                     | VARIABLE     |
| Aetna                 | VARIABLE (~90d in-net / 12mo OON per contract; 180d Medicaid MC) | Reconsideration ~180d → 2nd-level ~60d (verify)                                 | Availity / fax / mail                          | Medium       |
| Cigna                 | VARIABLE (~90d in / 180d OON; CA 365d)                           | Appeal within 180 calendar days (per Cigna appeals page)                        | CignaforHCP / payment-review form / mail / fax | Medium-High  |
| Humana                | MA 365d from DOS (fixed); commercial VARIABLE                    | MA reconsideration 65d (fixed); commercial ~180d                                | provider.humana.com                            | MA: High     |
| Elevance / Anthem     | UNKNOWN / VARIABLE by state (90 vs 180d cited)                   | Provider Dispute Resolution: 2-step, state-specific                             | Availity Payment Appeal Tool                   | Low/Variable |
| Florida Blue (BCBS)   | VARIABLE (~365d default; contract may be shorter)                | ~1yr most denials; MA 60d                                                       | Availity PASSPORT (Electronic Appeal)          | Low/Variable |
| Medicare FFS (CMS)    | 1 year from DOS (fixed rule)                                     | Redetermination 120d → Reconsideration 180d → ALJ 60d → Council 60d → court 60d | MAC                                            | High         |
| Medicare Advantage    | MA 365d claim floor                                              | Plan reconsideration 65d → IRE (C2C from 5/1/2026) → ALJ → Council → court      | Plan portal / CMS                              | High         |
| Medicaid managed care | State-set (fixed by 42 CFR 438)                                  | Plan appeal ≤30d resolution → state fair hearing (request 90–120d)              | Plan portal / state                            | High         |

CY2026 Medicare appeal amount-in-controversy thresholds: \$200 (ALJ), \$1,960 (federal court). The Part C IRE changed from Maximus to C2C Innovative Solutions effective May 1, 2026.

## 12. Coverage Policies

Coverage/medical-necessity policy is retrievable and, for CMS, highly machine-mappable. CMS NCDs and LCDs (with coding Articles) live in the free Medicare Coverage Database and map to CPT/HCPCS/ICD-10 — the richest free source for denial-prevention logic. Commercial medical policies (e.g., Aetna CPBs, UHC policies) are retrievable but non-uniform. InterQual (Optum) and MCG (Hearst) UM criteria are **license-gated** — budget for licensing, not scraping.

| Policy source                                      | Code-mappable                                 | Access                                                |
|----------------------------------------------------|-----------------------------------------------|-------------------------------------------------------|
| CMS NCDs (National Coverage Determinations)        | YES (CPT/HCPCS/ICD-10 via MCD)                | Free, public (Medicare Coverage Database)             |
| CMS LCDs + Articles (Local Coverage)               | YES (richest machine-mappable source)         | Free, public (MCD)                                    |
| Payer medical policies / Clinical Policy Bulletins | PARTIAL (CPT/HCPCS referenced; format varies) | Public per payer; not uniform                         |
| InterQual (Optum)                                  | YES (within product)                          | LICENSE-GATED — cannot scrape; commercial license/API |
| MCG (Hearst Health)                                | YES (within product)                          | LICENSE-GATED — commercial license                    |

### Boundary

AI can retrieve, parse and map coverage policy to codes for denial-prevention (provider-side) and decision-support (payer-side). It cannot be the sole basis for a final medical-necessity determination — a qualified clinician must make and own any adverse decision.

## 13. CMS, Medicare & Medicaid

The programs must not be conflated. Medicare FFS, Medicare Advantage, Medicaid (FFS and managed care), CHIP and QHPs each carry distinct rules, appeal ladders and interoperability obligations.

| Program               | Appeal ladder                                                                                                                     |
|-----------------------|-----------------------------------------------------------------------------------------------------------------------------------|
| Medicare FFS          | 5 levels: 1) Redetermination (MAC), 2) Reconsideration (QIC), 3) ALJ/OMHA hearing, 4) Medicare Appeals Council, 5) Federal court. |
| Medicare Advantage    | Organization determination → reconsideration → Independent Review Entity (IRE/Maximus) → ALJ → Council → court.                   |
| Medicaid managed care | Plan appeal (42 CFR 438 subpart F) → state fair hearing; external review where applicable.                                        |
| Commercial / ERISA    | Internal appeal(s) per 29 CFR 2560.503-1 → external review (IRO) under ACA (45 CFR 147.136).                                      |

Under **CMS-0057-F**, CMS-regulated payers (MA, Medicaid & CHIP, FFE QHPs) must implement FHIR Patient Access (enhanced), Provider Access, Payer-to-Payer and Prior Authorization APIs — API build-out generally due **Jan 1, 2027**; operational PA rules (72-hour expedited / 7-day standard; specific denial reason; public metrics) from **Jan 1, 2026**. It does not apply to commercial/employer (ERISA) plans and does not make eligibility API-only.



## 14. FHIR & EDI Standards

### 14.1 FHIR resources

| Resource / API                        | Rule       | Status      | RCM relevance                     |
|---------------------------------------|------------|-------------|-----------------------------------|
| Patient Access API                    | CMS-9115-F | Live (2021) | Member data — NOT provider RCM    |
| Provider Directory API                | CMS-9115-F | Live (2021) | Directory data                    |
| Provider Access API                   | CMS-0057-F | 2027        | RCM-relevant new API              |
| Payer-to-Payer API                    | CMS-0057-F | 2027        | Data follows patient              |
| Prior Authorization API               | CMS-0057-F | 2027        | Wraps X12 278                     |
| Coverage Requirements Discovery (CRD) | Da Vinci   | Emerging    | Denial prevention                 |
| Documentation Templates & Rules (DTR) | Da Vinci   | Emerging    | PA docs                           |
| Clinical Data Exchange (CDex)         | Da Vinci   | Emerging    | Attachments                       |
| CARIN Blue Button                     | CARIN      | Live        | Claims content for Patient Access |

### 14.2 EDI (X12) matrix

| X12   | Purpose                           | HIPAA                   | Clearinghouse dep. |
|-------|-----------------------------------|-------------------------|--------------------|
| 270   | Eligibility inquiry               | Mandated                | YES                |
| 271   | Eligibility response              | Mandated                | YES                |
| 276   | Claim status inquiry              | Mandated                | YES                |
| 277   | Claim status response             | Mandated                | YES                |
| 277CA | Claim acknowledgment              | Not mandated            | YES                |
| 278   | Services review / prior auth      | Mandated                | YES                |
| 837P  | Professional claim                | Mandated                | YES                |
| 837I  | Institutional claim               | Mandated                | YES                |
| 837D  | Dental claim                      | Mandated                | YES                |
| 835   | Remittance advice (ERA)           | Mandated                | YES                |
| 275   | Additional information/attachment | Not mandated (proposed) | PARTIAL            |
| 999   | Implementation acknowledgment     | Not mandated            | YES                |
| TA1   | Interchange acknowledgment        | Not mandated            | YES                |

### 14.3 Attachments

Claim/PA supporting documentation moves via the legacy **X12 275** ('Additional Information') or the emerging FHIR **Da Vinci CDex** (\$submit-attachment). There is no finalized federal claims-attachments mandate (status UNKNOWN); CDex adoption is early. AI can identify missing documents, classify and extract, associate to claims, and route — a strong near-term capability.

## 15. Regulatory Models & 'Billing on Behalf of Payers'

Doing claims/denial/UM/appeal work for payers spans a spectrum. Obligations attach to **who exercises discretion**: as a vendor moves from decision-support to delegated decision-making, it progressively becomes the regulated actor (TPA, UR agent, potential ERISA fiduciary, MA/Medicaid FDR/subcontractor).

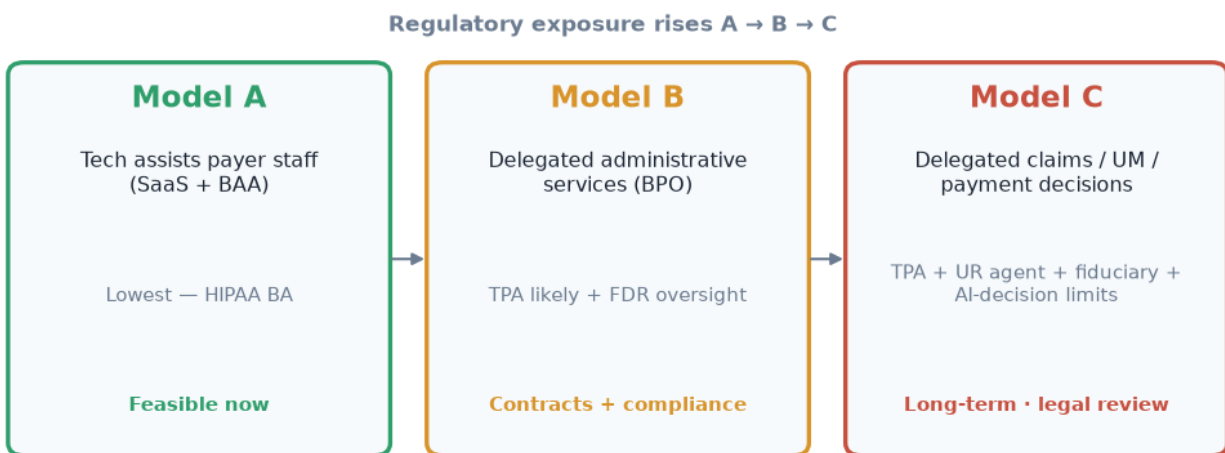


Figure 5. Three models of payer-side engagement — regulatory exposure rises A → B → C.

**Who the rules attach to.** Regulatory obligations attach to who exercises discretion. Moving Model A→B→C, the vendor progressively becomes the regulated actor (TPA / UR agent / potential ERISA fiduciary / FDR) rather than a tool supplier.

**TPA licensing.** Most states license/register Third-Party Administrators that adjust/settle/administer claims for plans (NAIC TPA model act). Pure SaaS (Model A) is usually not a TPA; adjudicating/processing claims (Model B) typically triggers TPA licensure. State-by-state — legal review required.

**Utilization management.** State UR laws + URAC/NCQA accreditation govern UM conduct/timeframes. Adverse medical-necessity determinations must be made by a qualified clinician (licensed physician/peer in specialty) — not AI alone. If the vendor renders determinations it becomes a UR agent.

**Medicare Advantage.** 42 CFR 422: MA org retains responsibility; contractors are first-tier/downstream/related entities (FDRs) subject to CMS oversight/audit. Adverse determinations need physician review (422.566(d)). CMS-4201-F: medical-necessity must be individualized, not based on an algorithm ignoring the patient — AI cannot be the sole basis.

**Medicaid managed care.** 42 CFR 438: MCO retains full responsibility; subcontractors (438.230) inherit oversight, 10-yr audit access, appeals-record retention; states may require prior approval.

**ERISA.** 29 CFR 2560.503-1 governs claims/appeals timeframes and 'full and fair review' for self-funded plans. A vendor exercising discretion over claims/appeals (Model C) may become an ERISA fiduciary. Decision-support (A) / ministerial (B) generally are not — fact-specific.

**HIPAA / BAA.** A vendor handling PHI is a business associate under a BAA in all models — but a BAA is a floor, not a ceiling. It governs PHI safeguards; it does not authorize the TPA/UR/fiduciary/FDR obligations that attach when the vendor makes decisions.

**External review / prompt pay.** State prompt-pay laws apply to anyone adjudicating/paying claims (B/C). ACA requires external review via independent IROs (45 CFR 147.136); vendors handling appeals must interoperate with — not substitute for — that layer.

**AI restriction trend.** Restrictions on AI in coverage/PA decisions are accelerating: CMS-4201-F (MA, individualized determinations) and California SB 1120 (eff. Jan 1, 2025 — final medical-necessity decisions by a licensed clinician, not AI). Multi-state, fast-moving — per-state legal review required.

**Not legal advice — review required**

Model A (SaaS decision-support under a BAA) is the startup-appropriate entry. Models B and C likely trigger state TPA licensing, UR/UM accreditation and clinician-review requirements, MA (42 CFR 422)/Medicaid (42 CFR 438) delegation and FDR oversight, and potential ERISA fiduciary status. AI cannot make final medical-necessity/coverage decisions alone (CMS-4201-F; CA SB 1120). Each payer-side model requires comprehensive regulatory/legal review.

## 16. Build / Buy / Partner

| Capability                        | Decision                   | Rationale                                                                                 |
|-----------------------------------|----------------------------|-------------------------------------------------------------------------------------------|
| Eligibility (270/271)             | BUY / PARTNER              | Clearinghouse API (Stedi/Optum/Availity); commoditized. Don't rebuild connectivity.       |
| Benefits (271 parsing)            | BUY + BUILD                | Buy the transaction; build structured benefit normalization.                              |
| Claim Status (276/277)            | BUY / PARTNER              | Clearinghouse API; commoditized.                                                          |
| Claim Submission (837)            | BUY / PARTNER              | Clearinghouse; add validation/scrubbing (BUILD).                                          |
| ERA/Denials (835)                 | BUY + BUILD                | Buy the 835; build denial intelligence (root cause, recovery).                            |
| Patient Responsibility Estimation | BUILD                      | Own estimation engine over payer data; label as estimate.                                 |
| Prior Authorization               | PARTNER + BUILD            | Portal/voice now; FHIR PAS (2027); build requirements/denial intelligence.                |
| Appeals                           | BUILD + HUMAN              | No standard API; build drafting/orchestration; human authors clinical appeals.            |
| AI Voice                          | BUY / INTEGRATE            | Voice provider/telephony; orchestrate as last resort.                                     |
| Payer Intelligence                | BUILD                      | Proprietary moat — payer behavior, channels, outcomes.                                    |
| Orchestration engine              | BUILD                      | Proprietary moat — right-channel decisioning + audit.                                     |
| Payment Integrity                 | BUILD (later) / PARTNER    | Large regulated market; enter after provider-side base.                                   |
| Payer-side Adjudication           | PARTNER / HIGHLY REGULATED | Decision support only; delegated functions need contract + oversight + legal review.      |
| Coverage-policy retrieval/parsing | BUILD                      | Retrieve NCD/LCD + payer medical policies; map to CPT/HCPCS/ICD-10 for denial prevention. |

# 17. Recommended Architecture

One Payer-Intelligence + Orchestration core serves both sides, selecting the cheapest sufficient channel per payer and workflow — EDI → API/FHIR → Portal → Voice → Human — with human exception handling and full audit. Provider-side capabilities ship first; payer-side capabilities layer on as decision-support (Model A) and administrative services (Model B).

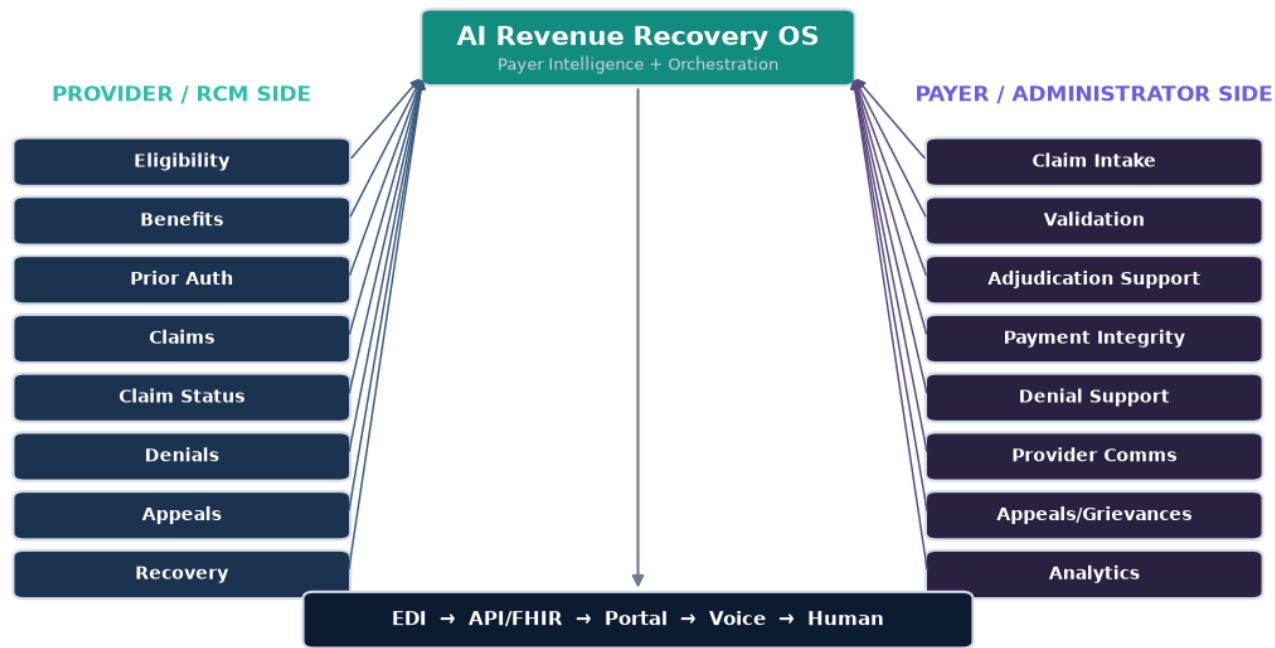


Figure 6. Two-sided architecture over existing connectivity, with the proprietary intelligence + orchestration layer as the moat.

# 18. Product Roadmap

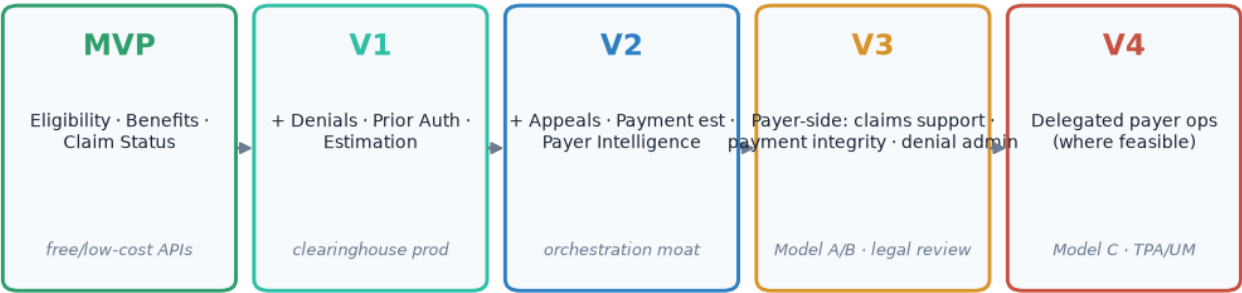


Figure 7. MVP → V4. Provider-side first on free/low-cost APIs; payer-side as decision-support; delegated payer operations only where legally/contractually feasible.

- **MVP:** Eligibility + Benefits + Claim Status (clearinghouse API; Stedi free tier → near-\$0).
- **V1:** + Denial intelligence (835), Prior-auth intelligence, Patient-responsibility estimation.
- **V2:** + Appeals orchestration, Payer Intelligence layer (the moat).
- **V3:** Payer-side claims-support / payment-integrity / denial-support as Model A decision-support (legal review).
- **V4:** Delegated payer operations (Model C) only where legally and contractually feasible.

## 19. Risks & Regulatory Considerations

- **Pricing opacity.** Exact clearinghouse per-transaction pricing is behind calculators/contracts — model economics only after capturing real numbers.
- **Payer-side regulation.** Models B/C likely require TPA licensing, UM accreditation and clinician review; multi-state and fact-specific. Legal review required before any delegated function.
- **AI-decision limits.** CMS-4201-F and CA SB 1120 (and a growing multi-state trend) bar AI-alone medical-necessity decisions — architect human-in-the-loop from day one.
- **Guideline variability.** Commercial timely-filing/appeal windows are contract/plan/state-variable — store as configurable values, never hard-coded.
- **Coverage-content licensing.** InterQual/MCG are license-gated; budget for licensing rather than scraping.
- **Competitive consolidation.** Payment integrity is consolidating (New Mountain, Cotiviti, Optum) — enter with a wedge, not breadth.
- **Coverage limitation.** 43 orgs are representative; several data points are UNKNOWN/VARIABLE pending per-payer/state verification (Research Gaps appendix).

## 20. Conclusion

AI Revenue Recovery OS can realistically **start as a provider/RCM-side platform** using existing healthcare APIs and clearinghouses — with a near-\$0 MVP — and **eventually become a payer-side claims intelligence and operations platform**. The provider side is a build-and-integrate problem largely solved by clearinghouse infrastructure; the durable asset is the Payer-Intelligence + Orchestration layer. The payer side is a large adjacent opportunity, best entered as decision-support and administrative services, with delegated decisioning reserved for a later, carefully-regulated stage.

### Final answer

**YES, WITH CONDITIONS.** Provider-side now on existing rails; payer-side as Model A/B decision-support and administrative services; Model C (delegated claims/UM/payment) long-term and only after comprehensive regulatory/legal review. Final medical-necessity/coverage decisions remain with qualified clinicians — AI assists, humans decide.



## Appendix A. Payer-Side Opportunity Scores

| Payer-side workflow                 | Score | Band   | Model               |
|-------------------------------------|-------|--------|---------------------|
| Claim Intake & Validation AI        | 76    | HIGH   | Model A (SaaS)      |
| Eligibility/Benefit Validation      | 72    | HIGH   | Model A             |
| Coding/Policy Checks (edits)        | 69    | MEDIUM | Model A/B           |
| Adjudication Decision Support       | 60    | MEDIUM | Model A → C         |
| Payment Integrity (prepay/postpay)  | 70    | HIGH   | Model A/B           |
| Denial Decision Support             | 60    | MEDIUM | Model A             |
| Provider Communication              | 71    | HIGH   | Model A/B           |
| Appeal/Grievance Workflow           | 56    | MEDIUM | Model B             |
| Utilization Mgmt / Prior Auth Admin | 54    | MEDIUM | Model C (delegated) |
| Delegated Claims Operations / TPA   | 50    | MEDIUM | Model C (TPA)       |

## Appendix B. Sources

| Source                                                                    | Type     | Confidence |
|---------------------------------------------------------------------------|----------|------------|
| CMS-0057-F Interoperability & Prior Authorization Final Rule — Fact Sheet | OFFICIAL | High       |
| CMS-0057-F Federal Register (89 FR 8758, doc 2024-00895)                  | OFFICIAL | High       |
| CMS Prior Authorization API FAQ                                           | OFFICIAL | High       |
| CMS Provider Access API FAQ                                               | OFFICIAL | High       |
| CMS-9115-F Interoperability and Patient Access — Fact Sheet               | OFFICIAL | High       |
| CMS-9115-F Payer-to-Payer enforcement discretion FAQ                      | OFFICIAL | High       |
| CMS Administrative Simplification — Adopted Standards & Operating Rules   | OFFICIAL | High       |
| CMS Health Care Transactions Basics                                       | OFFICIAL | High       |
| CMS Operating Rules for Eligibility and Claims Status                     | OFFICIAL | High       |
| CMS APIs & Implementation Guides (Standards/IGs)                          | OFFICIAL | High       |
| X12 Claim Adjustment Reason Codes (CARC)                                  | STANDARD | High       |
| X12 Remittance Advice Remark Codes (RARC)                                 | STANDARD | High       |
| CAQH Index Report 2024 — From Transactions to Trust                       | INDUSTRY | High       |
| CAQH 2024 Index Report — Key Takeaways                                    | INDUSTRY | High       |
| CAQH CORE Eligibility & Benefits (270/271) Data Content Rule              | STANDARD | High       |
| CAQH CORE Payment & Remittance (835/EFT) Reassociation Rule               | STANDARD | High       |
| CAQH CORE Prior Authorization White Paper (2024)                          | INDUSTRY | High       |
| HL7 Da Vinci Prior Authorization Support (PAS) IG                         | STANDARD | High       |
| HL7 Da Vinci Coverage Requirements Discovery (CRD) IG                     | STANDARD | High       |
| HL7 Da Vinci Documentation Templates and Rules (DTR) IG                   | STANDARD | High       |
| HL7 Da Vinci Clinical Data Exchange (CDex) IG                             | STANDARD | High       |
| HL7 Da Vinci Payer Data Exchange (PDex) IG                                | STANDARD | High       |
| CARIN Alliance IG for Blue Button (CARIN-BB)                              | STANDARD | High       |
| Stedi Healthcare docs                                                     | VENDOR   | High       |
| Stedi pricing (pay-as-you-go)                                             | VENDOR   | High       |
| Stedi Basic (free) plan — 100 free transactions/mo                        | VENDOR   | High       |
| Stedi pay-as-you-go pricing announcement                                  | VENDOR   | High       |
| Stedi payer network (3,500+ payers)                                       | VENDOR   | High       |
| Stedi transaction enrollment                                              | VENDOR   | High       |
| Availity API Marketplace                                                  | VENDOR   | High       |
| Availity HIPAA Transactions (developer blog)                              | VENDOR   | High       |
| Availity developer getting started (Demo vs Standard plan)                | VENDOR   | High       |
| Optum (Change Healthcare) developer portal                                | VENDOR   | High       |
| Optum Eligibility & Claims API — get started                              | VENDOR   | High       |
| Waystar platform — claim management                                       | VENDOR   | Medium     |
| Zelis provider payments / ERA                                             | VENDOR   | High       |
| pVerify eligibility/claim-status/PA API (FHIR)                            | VENDOR   | Medium     |
| Claim.MD clearinghouse API                                                | VENDOR   | Medium     |
| Office Ally EDI clearinghouse                                             | VENDOR   | Medium     |
| Inovalon (ABILITY) provider cloud / ONE developer portal                  | VENDOR   | Medium     |
| ONC Lantern FHIR endpoint monitoring                                      | OFFICIAL | High       |

| Source                                                                           | Type     | Confidence |
|----------------------------------------------------------------------------------|----------|------------|
| CARIN Alliance FHIR Directory Framework                                          | INDUSTRY | Medium     |
| CMS Blue Button 2.0 (Medicare FFS claims FHIR API)                               | OFFICIAL | High       |
| CMS Data at the Point of Care (DPC)                                              | OFFICIAL | High       |
| CMS AB2D (claims data for MA/Part D)                                             | OFFICIAL | Medium     |
| NPPES NPI Registry API                                                           | OFFICIAL | High       |
| Synthea synthetic patient generator                                              | STANDARD | High       |
| ONC Inferno / (g)(10) FHIR test kit                                              | OFFICIAL | High       |
| Logica Health FHIR sandbox                                                       | STANDARD | Medium     |
| CMS Medicare Coverage Database (NCD/LCD)                                         | OFFICIAL | High       |
| CMS Medicare Advantage appeals (Part C)                                          | OFFICIAL | High       |
| CMS Original Medicare (FFS) appeals — 5 levels                                   | OFFICIAL | High       |
| 42 CFR Part 422 (Medicare Advantage)                                             | OFFICIAL | High       |
| 42 CFR Part 438 (Medicaid managed care)                                          | OFFICIAL | High       |
| 29 CFR 2560.503-1 (ERISA claims procedure)                                       | OFFICIAL | High       |
| NAIC TPA (Registration & Regulation) model / state licensing                     | OFFICIAL | Medium     |
| URAC Health Utilization Management accreditation                                 | INDUSTRY | Medium     |
| NCQA Utilization Management accreditation                                        | INDUSTRY | Medium     |
| CMS first-tier, downstream & related entities (FDR) oversight                    | OFFICIAL | Medium     |
| HL7 Da Vinci CDex — attachments for claims/PA (FHIR)                             | STANDARD | High       |
| X12 275 — Additional Information / claim attachment                              | STANDARD | Medium     |
| CMS-4201-F (2024 MA & Part D final rule — AI/algorithm coverage-decision limits) | OFFICIAL | High       |
| California SB 1120 — Physicians Make Decisions Act (AI in UM, eff. Jan 1, 2025)  | OFFICIAL | High       |
| HHS HIPAA Business Associate guidance (45 CFR 160.103 / 164.504(e))              | OFFICIAL | High       |
| ACA external review / IRO (45 CFR 147.136)                                       | OFFICIAL | High       |
| Cotiviti payment accuracy                                                        | VENDOR   | High       |
| Optum payment integrity                                                          | VENDOR   | High       |
| Zelis payment integrity / ZIPP                                                   | VENDOR   | High       |
| Cohere Health (payer UM/PA AI)                                                   | VENDOR   | High       |
| Anterior (payer clinical decisioning AI)                                         | VENDOR   | High       |
| Da Vinci CDex attachments (\$submit-attachment)                                  | STANDARD | High       |
| UnitedHealthcare Care Provider Administrative Guide                              | OFFICIAL | Medium     |
| Aetna Clinical Policy Bulletins (CPBs)                                           | OFFICIAL | Medium     |
| Cigna appeals & disputes (180-day appeal)                                        | OFFICIAL | High       |
| Humana reconsiderations & appeals                                                | OFFICIAL | Medium     |
| Anthem/Elevance provider news & policies (via Availity)                          | OFFICIAL | Low        |
| Florida Blue provider (appeals via Availity PASSPORT)                            | OFFICIAL | Low        |
| CMS Medicare appeals AIC thresholds CY2026 (Federal Register)                    | OFFICIAL | High       |
| Optum InterQual clinical criteria (license-gated)                                | VENDOR   | High       |

Full URLs are hyperlinked in the Excel SOURCES sheet. Primary hubs: cms.gov, ecf.gov (Titles 42/29/45), x12.org, hl7.org, caqh.org, naic.org, and vendor developer/pricing portals.

## Appendix C. Research Gaps

Explicitly enumerated coverage limitations requiring primary-source or per-payer/state verification before production commitments:

- Exact clearinghouse per-transaction pricing (Stedi calculator; Optum/Availity quotes).
- Commercial timely-filing / appeal deadlines — contract/plan/state-variable; UHC/Anthem have no single value.
- Payment-integrity market size — firm estimates vary widely; cite a range.
- Payer-side regulatory specifics (TPA/UM/ERISA) — state-by-state; legal review required.
- FHIR versions & Provider Access readiness — R4 implied; 2027 APIs not yet live.
- InterQual/MCG licensing terms — proprietary, not public.
- DPC production paused; Logica retiring to Meld — verify current status.
- Full payer universe — 43 orgs representative, not exhaustive (thousands of payer IDs).

**Research coverage limitation.** This is a feasibility analysis from public documentation, not legal advice. Payer-side models B and C, and any delegated function, require comprehensive regulatory/legal review before commitment.